

Code:

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/* Toufiq Islam - 2310728141 */
//determinant
#include<iostream>
#include<cmath>
#include<iomanip>
using namespace std;
int main()
{
    int n,i,j,k;
    double t;
    cout<<"Please enter the dimension of the matrix : \n";
    cin>>n;
    double mat[n][n], determinant=1.0;

    cout<<"\n Please enter the matrix row wise : \n";
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            cin>>mat[i][j];

    cout<<"\n The matrix you have entered : \n";
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
            cout<<mat[i][j]<<" ";

        cout<<endl;
    }

    //pivot
    for(i=0;i<n;i++) //current row
        for(k=i+1;k<n;k++) //below the current row
            if(abs(mat[i][i])<abs(mat[k][i])) // ii means corner term and ki means same next row same column,note dekho.
            {
                for(j=0;j<n;j++) //across all coulmn
                {
                    double temp=mat[i][j];
                    mat[i][j]=mat[k][j];
                    mat[k][j]=temp;
                }
                determinant*=-1;
            }

    cout<<"\n The matrix after pivotising = \n";
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
            cout<<mat[i][j]<<" ";
        cout<<endl;
    }

    //gauss
    for(i=0;i<n-1;i++) //current row
        for(k=i+1;k<n;k++) //below the current row
        {
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        double t=mat[k][i]/mat[i][i];
        for(j=0;j<n;j++) //across all coulumn
            mat[k][j]=mat[k][j]-t*mat[i][j];
    }

    cout<<"\n The matrix after gauss = \n";
    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
            cout<<mat[i][j]<<" ";
        cout<<endl;
    }

    for(i=0;i<n;i++)
        determinant*=mat[i][i];
    cout<<"\n The required determinant is : \n";
    cout<<determinant;
    return 0;
}

```

Output:

Please enter the dimension of the matrix :

3

Please enter the matrix row wise :

2 -1 3

1 4 2

3 0 -2

The matrix you have entered :

2 -1 3

1 4 2

3 0 -2

The matrix after pivotising =

3 0 -2

1 4 2

2 -1 3

The matrix after gauss =

3 0 -2

0 4 2.66667

0 0 5

The required determinant is :

-60

Process exited after 54.23 seconds with return value 0

Press any key to continue . . .